

When Methodology Becomes the Problem: The Dark Side of Existing Software Engineering Methodologies in Financial Services

By Neeraj Aggarwal

Methodologies Become Dogma Instead of Tools

Teams often treat Agile, Scrum, SAFe, DevOps, or Waterfall as **religion**, not **strategy**.

- “Follow the process” replaces “solve the problem.”
- Rituals (standups, retros, PI planning) become performative.
- Teams optimize for **compliance**, not **outcomes**.

This is one of the biggest modernization killers you and I have discussed before — process rigidity masquerading as discipline.

They Assume Ideal Conditions That Rarely Exist

Most methodologies assume:

- Stable teams
- Clear requirements
- Low technical debt
- High engineering maturity
- Strong product ownership

Real enterprises — especially banks — have **none** of these.

So the methodology collapses under the weight of reality.

They Ignore Legacy Complexity

Traditional methodologies were never designed for:

- 40-year-old COBOL systems
- Missing documentation
- Tribal knowledge
- Interdependent batch cycles
- Regulatory constraints

This is why modernization programs fail even when “Agile” is followed perfectly.

The methodology simply doesn’t match the **architecture reality**.

They Create Illusion of Progress (Velocity Theater)

Teams deliver:

- Story points
- Burndown charts
- Sprint demos

...but not **actual business value**.

Velocity becomes a vanity metric.

Teams “complete” work that doesn’t move the needle.

This is one of the most toxic dark sides of Agile in large enterprises.

They Overemphasize Process, Underemphasize Architecture

Most methodologies:

- Treat architecture as a one-time activity
- Assume modularity that doesn’t exist
- Ignore technical debt
- Underestimate refactoring needs

This leads to:

- Feature delivery on top of unstable foundations
- Accumulating debt
- Fragile systems
- Modernization paralysis

They Fail in Multi-Vendor, Multi-Team Environments

Methodologies assume:

- One team
- One backlog
- One product owner

But real enterprises have:

- 20+ teams
- 5 vendors
- Conflicting priorities
- Siloed ownership

Coordination overhead destroys the methodology's intent.

They Pretend Humans Are Rational and Predictable

Most methodologies assume:

- Perfect communication
- Honest estimation
- Stable morale
- No politics
- No fear of blame

But real teams have:

- Fear of escalation
- Hidden dependencies
- Conflicting incentives
- Burnout
- Skill gaps

Methodologies rarely account for **human factors**, which are the real drivers of success or failure.

They Don't Scale Well (Especially Agile)

Agile works beautifully for:

- Small teams
- Greenfield products
- Startups

But it breaks down in:

- Regulated industries
- Legacy modernization
- Multi-year programs
- Distributed teams

SAFe tries to fix this but often becomes **bureaucracy at scale**.

They Create Documentation Blind Spots

Many methodologies overcorrect:

- Waterfall → too much documentation

- Agile → too little documentation

In modernization, lack of documentation is catastrophic.

They Encourage Short-Term Delivery Over Long-Term Health

Most methodologies reward:

- Shipping features
- Closing sprints
- Hitting deadlines

They do **not** reward:

- Refactoring
- Reducing technical debt
- Improving architecture
- Stabilizing systems

This leads to long-term decay.

About Author

Neeraj Aggarwal is a recognized modernization leader with deep expertise in AI, core banking, and legacy transformation. He advises financial institutions on building resilient, data-driven architectures and contributes thought leadership across industry forums and publications.